BURMESE PRAGMATICS

NLP CLASSIFICATION SYSTEM

A 4-STAGE ABLATION STUDY ON CONTEXT-SENSITIVE PRAGMATIC CLASSIFICATION IN BURMESE

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QUICK RECAP

STAGE 1	STAGE 2	STAGE 3	STAGE 4
UTTERANCE ONLY	+ CONTEXT	ORACLE METADATA	PREDICTED METADATA
MACRO F1	MACRO F1	MACRO F1	MACRO F1
0.701	0.758	0.799	0.702

Key Finding: Stage 4 ≈ Stage 1 (0.702 ≈ 0.701) despite using predicted metadata. Error propagation from imperfect classifiers (A: 0.705, B: 0.598, C: 0.613) negates the oracle gain of +0.097.

RQ: CAN WE CLOSE THE GAP? → THAT'S WHAT THIS PRESENTATION ANSWERS

THREE NEW EXPERIMENTS

01 Codependent Chain

- RETRAIN B WITH [REGISTER] PREFIX
- RETRAIN C WITH [REGISTER] [POWER] PREFIX
- TEACHER FORCING AT TRAIN TIME
- CHAIN A→B_V2→C_V2 AT INFERENCE
- B: 0.598→0.637 C: 0.613→0.621

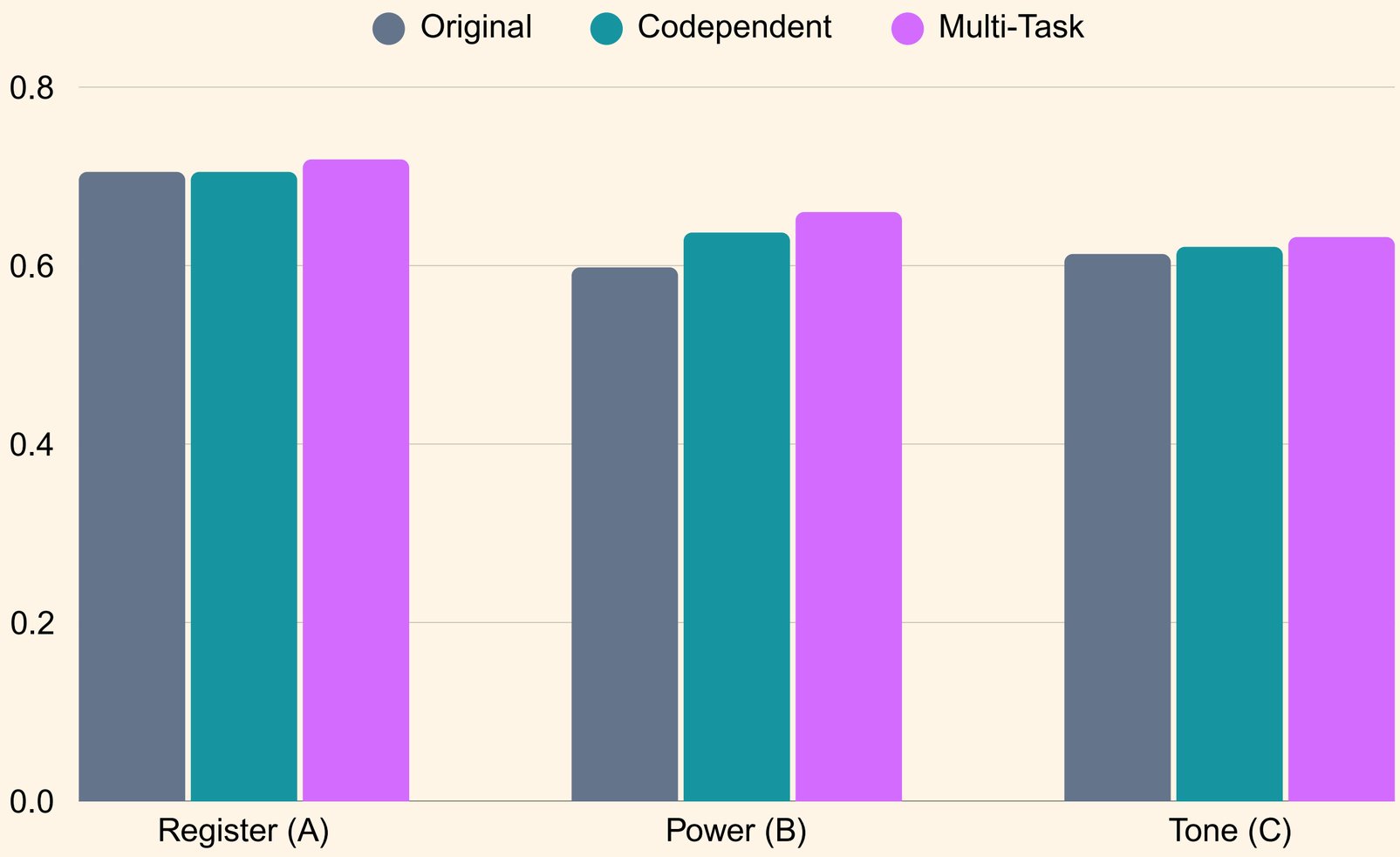
02 Multi-Task Classifier

- SINGLE XLM-R BACKBONE
- THREE INDEPENDENT HEADS (A+B+C)
- JOINT TRAINING WITH EQUAL LOSS WEIGHTING
- ONE FORWARD PASS AT INFERENCE
- B: 0.598→0.660 C: 0.613→0.632

03 Zero-Shot LLM Eval

- SAME 330 TEST EXAMPLES
- ORACLE METADATA PROVIDED (FAIR COMPARISON)
- CHATGPT O4-MINI: MACRO F1 0.635
- GEMINI 3 FAST: MACRO F1 0.493
- FINE-TUNED STAGE 1 BEATS BOTH LLMS

INTERMEDIATE CLASSIFIER IMPROVEMENT



Codependent

+0.039

Power (B) improvement

Multi-Task

+0.062

Power (B) improvement

Best overall

MT wins

all 3 heads improved

THE SURPRISING RESULT

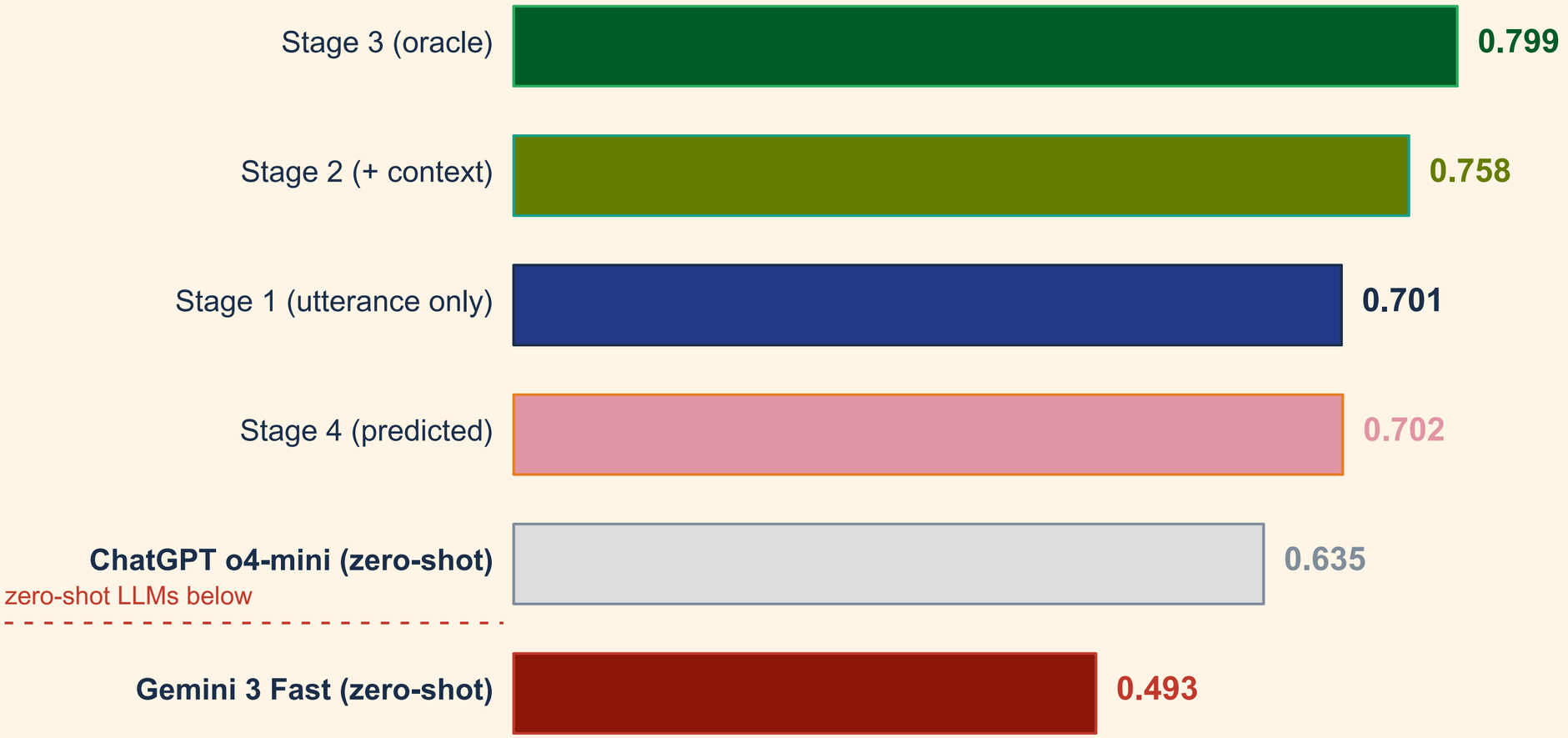
BETTER INTERMEDIATE CLASSIFIERS DID NOT IMPROVE END-TO-END PERFORMANCE

Stage 4 Original (independent)	Stage 4 v2 (codependent chain)	Stage 4 MT (multi-task)
0.702	**0.696**	**0.681**
Best	-0.006	-0.021

↑ Intermediate classifier quality

Why? Stage 3 politeness model was trained on oracle (ground-truth) metadata prefixes. Better predicted prefixes have **different error patterns** that Stage 3 handles worse. The original classifiers' errors happen to be most compatible with what Stage 3 trained on.

ZERO-SHOT LLM COMPARISON



Stage 1 (no metadata, no context) outperforms ChatGPT with full oracle metadata by +0.066 macro F1

FULL RESULTS

MODEL	ACCURACY	MACRO F1	NOTES
Stage 1 - Utterance only	0.739	0.701	Baseline
Stage 2 - + Context	0.794	0.758	+0.057 over Stage 1
Stage 3 - Oracle metadata	**0.846**	**0.799**	**★ Best | Oracle upper bound**
Stage 4 — Predicted (original)	0.727	0.702	≈ Stage 1 despite metadata
Stage 4 v2 — Codependent Chain	0.724	0.696	Better intermediate, worse end-to-end
Stage 4 MT — Multi-Task	0.773	0.681	Best intermediate, worst end-to-end
ChatGPT o4-mini (zero-shot)	0.564	0.635	*With oracle metadata*
Gemini 3 Fast (zero-shot)	0.500	0.493	*With oracle metadata*

ERROR ANALYSIS

49 CASES WHERE STAGE 3 IS CORRECT BUT STAGE 4 IS WRONG

METADATA ERROR RATES

Register errors

Raw rate: 26.1% → In failures: 63.3% of failures

Tone errors

Raw rate: 33.0% → In failures: 61.2% of failures

Power errors

Raw rate: 26.1% → In failures: 46.9% of failures

TOP POLITENESS CONFUSIONS

neutral → **polite** (21 cases)

register colloquial→standard

neutral → **informal** (7 cases)

tone neutral→emotional

informal → **polite** (5 cases)

register slang→colloquial

neutral → **rude** (5 cases)

tone neutral→negative

informal → **neutral** (5 cases)

register colloquial→standard

WHY LLMS FAIL ON BURMESE PRAGMATICS

CLASS	STAGE 3	STAGE 4	CHATGPT	GEMINI
Blunt	0.67	0.70	0.47	0.42
Informal	0.83	0.71	0.59	0.51
Neutral ★	0.87	0.77	0.45	0.54
Polite	0.82	0.67	0.64	0.36
Professional	0.94	0.91	0.79	0.52
Rude	0.67	0.46	0.87	0.60

NEUTRAL CLASS COLLAPSE

ChatGPT predicts neutral only 50× vs 174 true. Gemini 103× vs 174 true. LLMs conflate pragmatic neutrality with informal register.

INFORMAL OVER-PREDICTION

ChatGPT over-predicts informal by +72, Gemini by +61. 'Neutral → informal' is the #1 confusion for both LLMs.

RUDE EXCEPTION

ChatGPT achieves 0.87 F1 on rude — higher than Stage 3 (0.67). Explicit rudeness markers may transfer across languages.

KEY FINDINGS

1 CONTEXT CARRIES REAL SIGNAL

Adding social context improves macro F1 by +0.057. Burmese politeness cannot be classified from text alone — speaker relationship and situation matter.

2 ORACLE METADATA IS VALUABLE

Perfect metadata adds +0.041. Stage 3 achieves 0.799 macro F1 — a clear upper bound confirming that register, power, and tone are genuinely predictive.

3 BETTER CLASSIFIERS ≠ BETTER PIPELINE

Multi-task improves intermediate F1 by up to +0.062 but Stage 4 MT (0.681) is worse than original Stage 4 (0.702). Distribution mismatch is the bottleneck.

4 REGISTER IS THE CRITICAL METADATA

Register errors appear in 63.3% of Stage 3→Stage 4 failures. The colloquial→standard upgrade is the single most damaging error pattern.

5 FINE-TUNING BEATS ZERO-SHOT LLMS

Stage 1 (utterance only, no metadata) outperforms ChatGPT o4-mini with full oracle metadata by +0.066 macro F1. Domain fine-tuning is essential for low-resource pragmatics.

CONCLUSION

WHAT WE FOUND

✓ 4-stage ablation from utterance-only to full end-to-end pipeline

✓ Stage 3 (oracle) achieves 0.799 macro F1 — best result

✓ Error propagation costs −0.097 macro F1

✓ Improving intermediate classifiers does NOT help end-to-end

✓ Fine-tuned Stage 1 outperforms ChatGPT + Gemini with oracle metadata

✓ Register misclassification is the dominant failure mode

FUTURE WORK

→ Co-train politeness model jointly with metadata predictors

→ Explore Burmese-specific pre-trained encoders

→ Evaluate with more LLMs (Qwen, Grok, Claude)

→ Cross-validation for stable error propagation estimates

→ Formal annotation study for tone taxonomy validation

ALL MODELS ARE LIVE ON HUGGINGFACE

- ANNASUS10

DEMO:

BURMESE-PRAGMATICS-DEMO-V2

THANK YOU